

FUNCTIONAL TUMOR SUBCLONES PREDICT CLINICAL OUTCOMES IN MERKEL CELL CARCINOMA FROM BULK AND SINGLE-CELL RNA-SEQ

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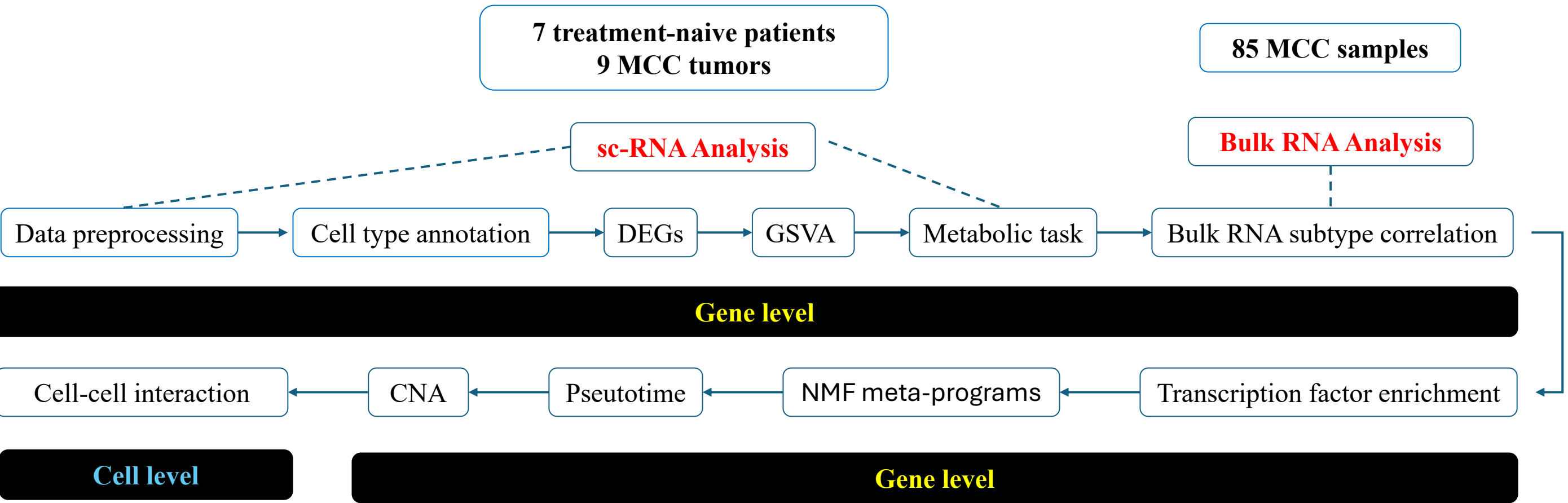
BACKGROUND

Merkel cell carcinoma (MCC) is a rare but aggressive skin cancer exhibiting molecular heterogeneity and variable clinical outcomes. In this study, we identified hallmark-inferred functional subtypes from bulk and sc-RNA sequencing data to stratify patients and predict prognosis.

MATERIALS AND METHODS

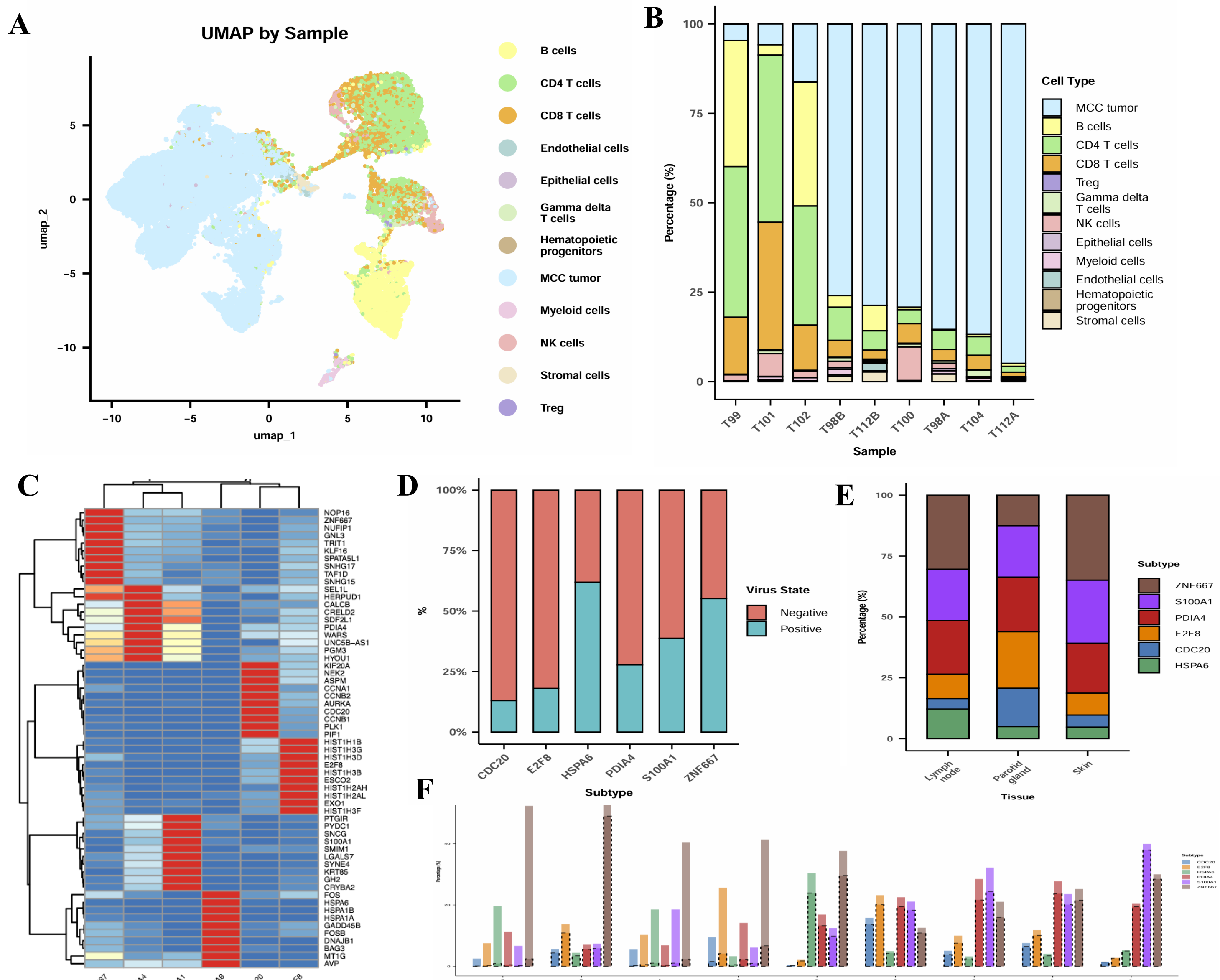
We constructed a single-cell transcriptomic landscape from 9 MCC tumor samples collected from lymph node metastases, primary skin lesions, and the parotid gland to characterize the cellular basis of subtype-specific functional programs.

WORKFLOW



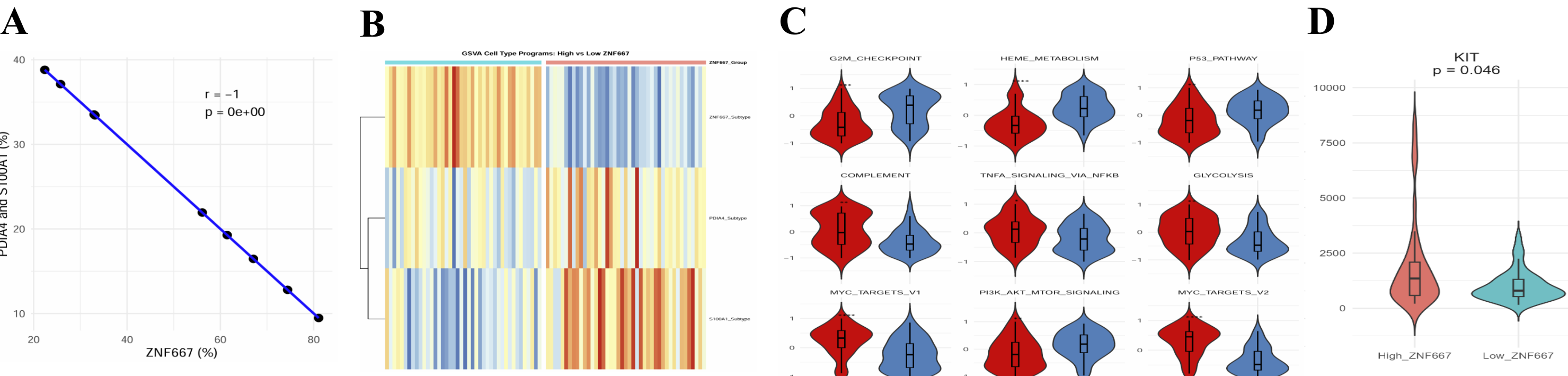
RESULTS

1. Single-cell atlas reveals MCC tumor subtypes with distinct tissue origins and viral associations



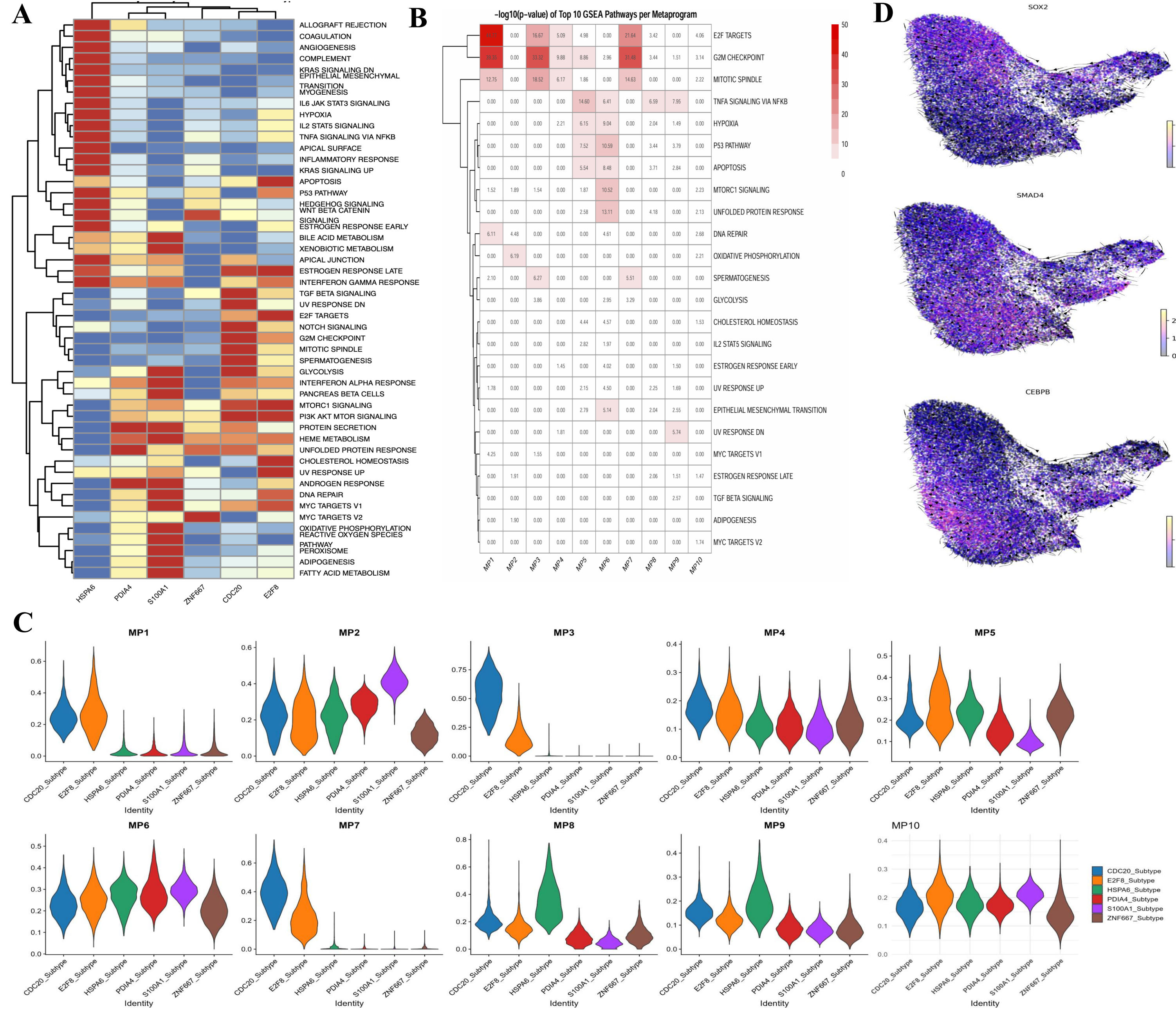
A) UMAP projection illustrates the cellular heterogeneity of MCC tumors, revealing distinct cell types distributed across the tumor microenvironment. **B)** Overall cell type composition shows the relative abundance of MCC_tumor (merged subtypes) and other cell types across individual samples. **C)** Differential gene expression (DEG) analysis highlights subtype-specific marker genes, revealing distinct molecular signatures among MCC subpopulations. **D)** Tumor subtypes show tissue-specific origins, as visualized in the stacked bar plot, highlighting enrichment patterns of subtypes across skin, lymph node, and distant metastases. **E)** Viral status distribution reveals subtype-specific associations with MCPyV infection. **F)** Subtype distribution across individual samples demonstrates inter-tumoral heterogeneity and supports stratification based on transcriptomic profiles.

3. Inverse regulation of ZNF667 and PDIA4/S100A1 defines prognostic subgroups in MCC tumors



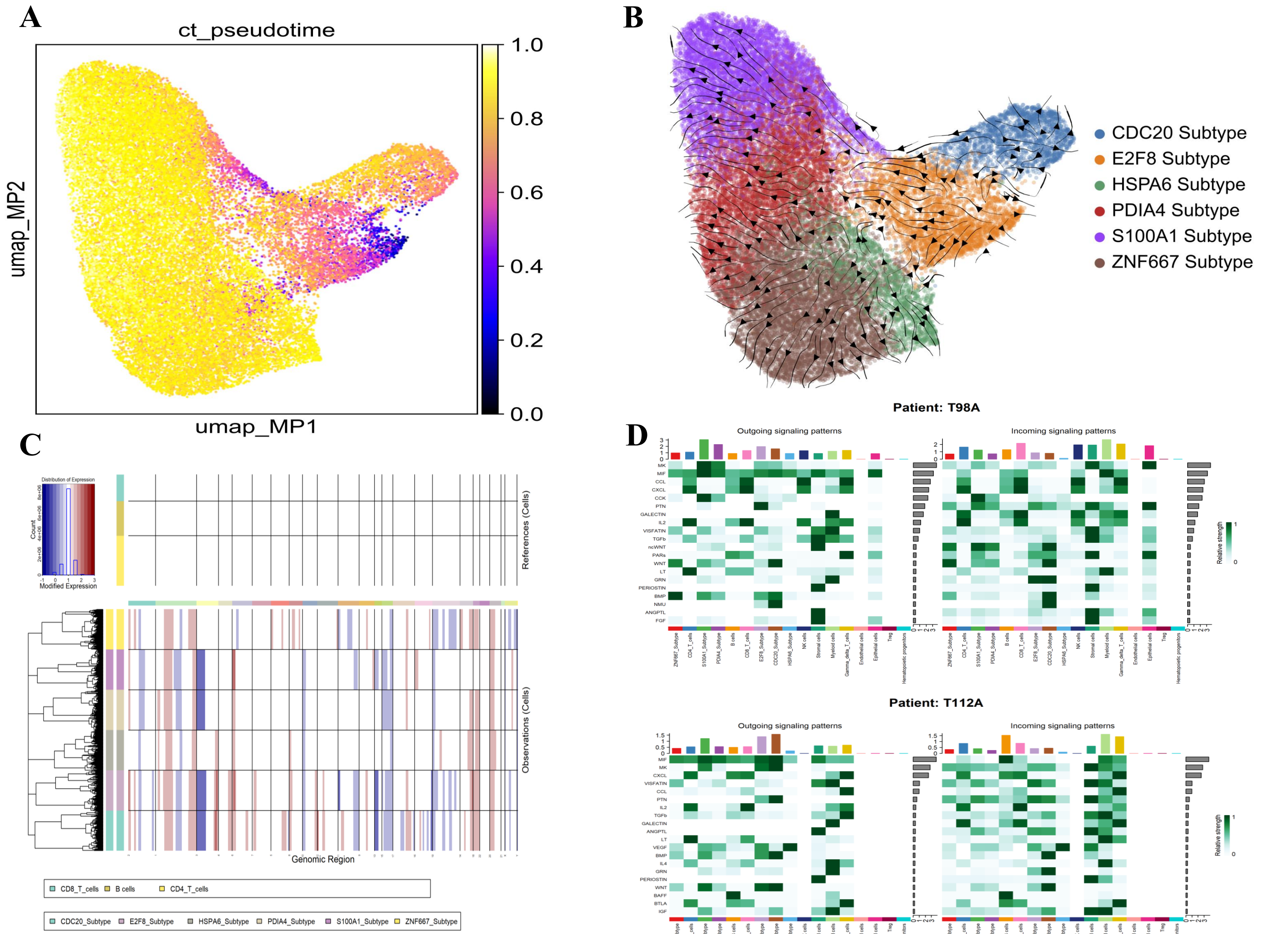
A) Inverse correlation between ZNF667 expression and combined PDIA4/S100A1 levels in MCC tumor cells. **B)** Bulk RNA-seq classification based on ZNF667 and PDIA4/S100A1 signatures identifies two molecular subgroups. **C)** Hallmark pathway activities between ZNF667-high and ZNF667-low subgroups. **D)** Higher KIT expression in ZNF667-high subgroup suggests association with poor survival

2. Distinct functional programs and regulatory factors define each tumor subtype



A) GSVA heatmap of hallmark pathways shows the global enrichment patterns across tumor subtypes. **B)** Heatmap of GSVA scores depict hallmark pathways enriched within each metaprogram, linking transcriptional modules to functional phenotypes. **C)** Violin plots illustrate metaprogram activity scores across subtypes, reflecting core transcriptional states in individual cells. **D)** UMAP feature plots showing expression of SOX2, SMAD4 and CEBPB across MCC tumor cells

4. Bulk-level molecular heterogeneity inferred from single-cell-derived cell-type and pathway signatures



A) Pseudotime reveals a continuum of cellular states among MCC tumor cells, suggesting a gradual transition across subtypes. Cells are colored by pseudotime scores. **B)** Inferred subtype evolution trajectory outlines potential progression paths among MCC subtypes, highlighting transitional states and branching points. **C)** Copy number variation (CNV) profiles inferred from single-cell transcriptomes reveal subtype-specific genomic alterations, with T and B cells as references for diploid baseline. **D)** Cell-cell communication analysis reveals subtype-specific signaling, with distinct outgoing strengths across tumor subtypes.

Conclusion

MCC subtypes represent a continuum from highly proliferative (E2F8, CDC20) to stress-adapted (HSPA6), metabolically active (PDIA4, S100A1), and quiescent (ZNF667) states, each with distinct communication profiles. These patterns reflect tumor evolution, microenvironmental adaptation, and potential therapeutic vulnerabilities

References

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